Subject Code: BCE302

PAPER ID-310437

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM III) THEORY EXAMINATION 2025-26
SURVEYING AND GEOMATICS

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 07 = 14

Q no.	Question	CO
a.	Mention two methods of ranging.	1
b.	Define degree of curve and mention its use in highway or railway alignment.	2
c.	Mention two advantages of Total Station.	3
d.	In a GIS raster model, each pixel represents an area of 25 m × 25 m. If a feature occupies 200 pixels, calculate its area.	3
e.	Distinguish between aerial photogrammetry and terrestrial photogrammetry	4
f.	What is spatial resolution in remote sensing?	4
g.	What is accuracy assessment in digital image analysis?	5

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

Q no.	Question	CO
a.	Describe meridians and bearings and their importance in surveying.	1
b.	A simple circular curve of radius $R = 400\ m$ is required to connect two straights with deflection angle $\Delta = 50^{O}$, Find: Tangent length (T), Length of curve (L), Length of long chord (C)	2
c.	Explain the principle of distance measurement using electromagnetic waves and describe any two types of Electronic Distance Measurement (EDM) instruments.	3
d.	Describe atmospheric windows and their importance in remote sensing.	5
e.	Explain UAV-based surveying. Describe its platform types, sensors used, and applications in civil engineering projects.	5

SECTION C

3. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	A baseline of 300 m was measured using a tape of 30 m nominal length. The tape was found to be 0.05 m too long. (i) Calculate the correct length of the baseline. (ii) If the measured angle subtended at the baseline is 45°, calculate the corrected length of the opposite side of the triangle.	1
b.	Explain reciprocal leveling. Discuss its necessity and sources of error eliminated by this method.	1

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4. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Explain the elements of a simple circular curve with neat sketch.	2
b.	Explain the importance of transition curves in highways and railways.	2

5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Discuss applications of GIS in civil engineering transportation planning and water resources.	3
b.	An EDM instrument measures the phase difference between transmitted and received signals. Wavelength of transmitted signal = 20 m Measured phase difference = 90°, Number of complete wavelengths = 150. Calculate the total distance measured.	3

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Describe photogrammetric mapping and its applications.	4
b.	Two photographs of the same area are taken from different positions. Parallax difference measured p = 2.5mm, flying height above ground H= 1200m, Focal length f = 150 mm, Air base =300m. Calculate the height of the object.	4

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO
a.	Write short notes on advantages and limitations of remote sensing in engineering projects.	5
b.	Explain different types of resolutions in remote sensing with suitable examples.	5

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